Methodology Passport Card

1. Project Information

- **Title:** Main factors affecting football players' market value in Big-5 European leagues: an econometric analysis by position groups
- **Authors:** Mukhamedzhan Kadir, Yerkanat Manapuly
- **GitHub Repository Link:** https://github.com/muhamed434/big5-player-valuation-econometrics

2. Type of Research

- **Type:** Quantitative Research (Econometric Analysis).
- **Justification:** The study relies on secondary quantitative statistical data collected from Transfermarkt and Fbref/StatsBomb. Econometric modeling (Pooled OLS, Fixed Effects, polynomial age terms) is applied to empirically test formal hypotheses regarding market value determinants across player positions.

3. Hypotheses Formulation

Research Question 1 (RQ1: Positional Valuation Differences)

- **Null Hypothesis (H0):** Controlling for playing time, league, and contract terms, the average log market value of players does not differ significantly across position groups (defenders, midfielders, forwards).
- **Alternative Hypothesis (H1):** Holding playing time, league, club status, and remaining contract length constant, forwards have a statistically significantly higher average log market value compared to defenders and midfielders.

Research Question 2 (RQ2: Non-linear Age Impact and Positional Peaks)

- **Null Hypothesis (H0):** Age has a strictly linear effect on market value, and the age of reaching peak market value is identical across all position groups.
- **Alternative Hypothesis (H1.1 - Non-linearity):** The effect of age on market value is non-linear (parabolic), where market value increases with age up to a specific peak, after which it declines.
- **Alternative Hypothesis (H1.2 - Positional Peak Differences):** The peak market value for defenders is reached at an older age (26 to 28 years old) than for attacking players (24 to 26 years old).

Research Question 3 (RQ3: Position-Specific Performance Metrics)

- **Null Hypothesis (H0):** Specialized defensive performance metrics have no statistically significant impact on the market value of defenders.
- **Alternative Hypothesis (H1):** Position-specific defensive metrics (tackles per 90, interceptions per 90, duel win percentage) explain defenders' market value better and with higher explanatory power (Adjusted R-squared) than standard aggregate "goals plus assists" metrics.

4. Matrix of Variables

Category	Variable Name	Description & Measurement Unit
Dependent Variable	log_market_value	Natural logarithm of player's market value in EUR: ln(Market Value)
Independent Variables	position	Categorical position group (Defender, Midfielder, Forward)
	age, age^2	Player age in years and quadratic age (Age squared) to model non-linear trajectories
	position_x_age	Interaction terms between age and position group
	tackles_per90	Successful tackles per 90 minutes (Defenders)
	interceptions_per90	Interceptions per 90 minutes (Defenders)
	duel_win_pct	Ground and aerial duel win percentage (%) (Defenders)
	prog_passes_per90	Progressive passes per 90 minutes (Midfielders)
	xg_per90	Non-penalty Expected Goals (xG) per 90 minutes (Forwards)
Control Variables	league	League fixed effects (EPL, La Liga, Serie A, Bundesliga, Ligue 1)

	minutes_played	Total official playing minutes in the season
	club_tier	Club financial status / league table finish tier
	contract_years	Remaining contract length in years

5. Selected Metrics and Baseline

Primary Metrics:

1. **Adjusted R-squared:** Main metric to measure model explanatory power across position-specific regressions.
2. **Statistical Significance (p-value < 0.05):** Evaluation of t-statistics to confirm or reject specific hypotheses.
3. **RMSE (Root Mean Squared Error):** Predictive accuracy of the log-transformed valuation models.

Guardrail Metrics:

1. **Multicollinearity Check (VIF < 5.0):** Variance Inflation Factor to ensure age and performance metrics do not distort standard errors.
2. **Heteroskedasticity Test (Breusch-Pagan p > 0.05):** Diagnostic test to verify constant residual variance.
3. **Sample Size Constraint (N >= 300):** Minimum required observations per positional subset.

Baseline Model:

- **Description:** A standard linear Pooled OLS model estimated on the aggregated player dataset without positional breakdown or specialized xG/defensive metrics. It includes only linear age, playing minutes, and simple "goals + assists per 90".